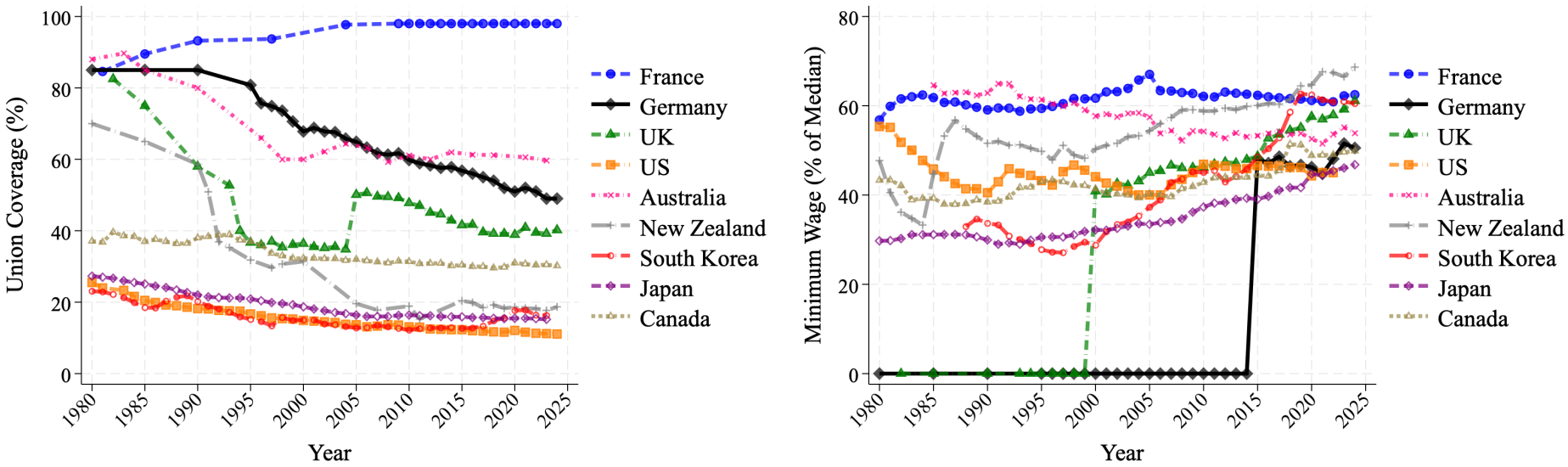
The Effects of The Minimum Wage on Collective Bargaining: Evidence from Germany

Alice Kügler[1,4,5] David Zentler-Munro[2,5,6] Nidhaan Jain[3]

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Workshop on Minimum Wages and Collective Bargaining, Mannheim, October 13 2026

Minimum Wage and Collective Bargaining Coverage Trends



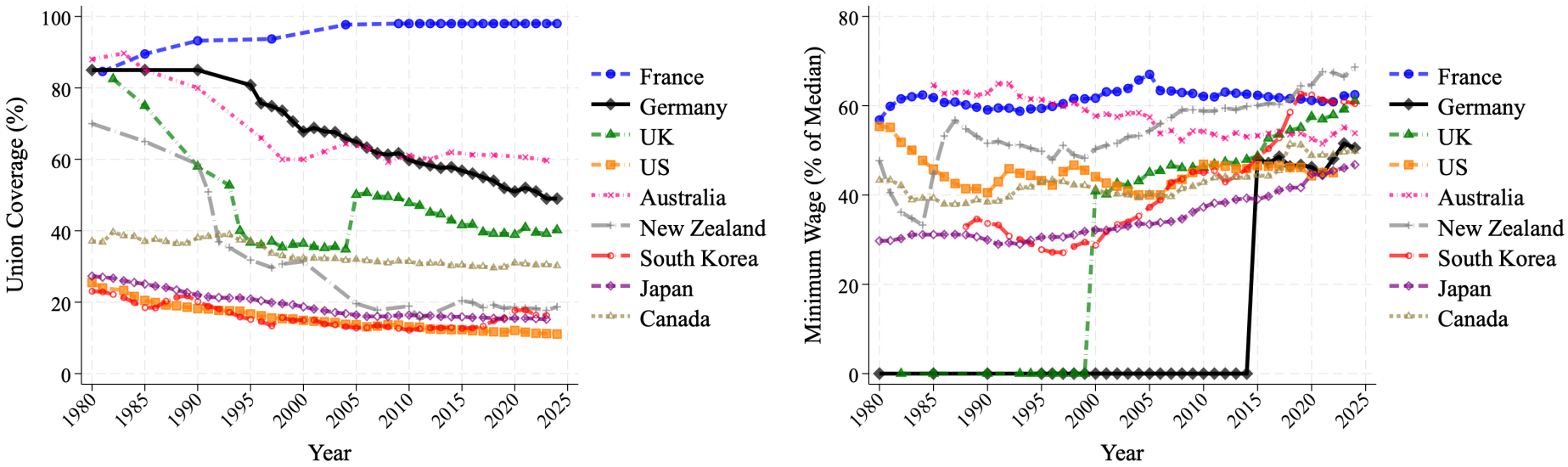
(a) Collective Bargaining Coverage

(b) Minimum Wage as % of Median

Source: OECD and authors' own calculations

- Simultaneous decline in collective bargaining coverage and increased reliance on minimum wages over time

Minimum Wage and Collective Bargaining Coverage Trends



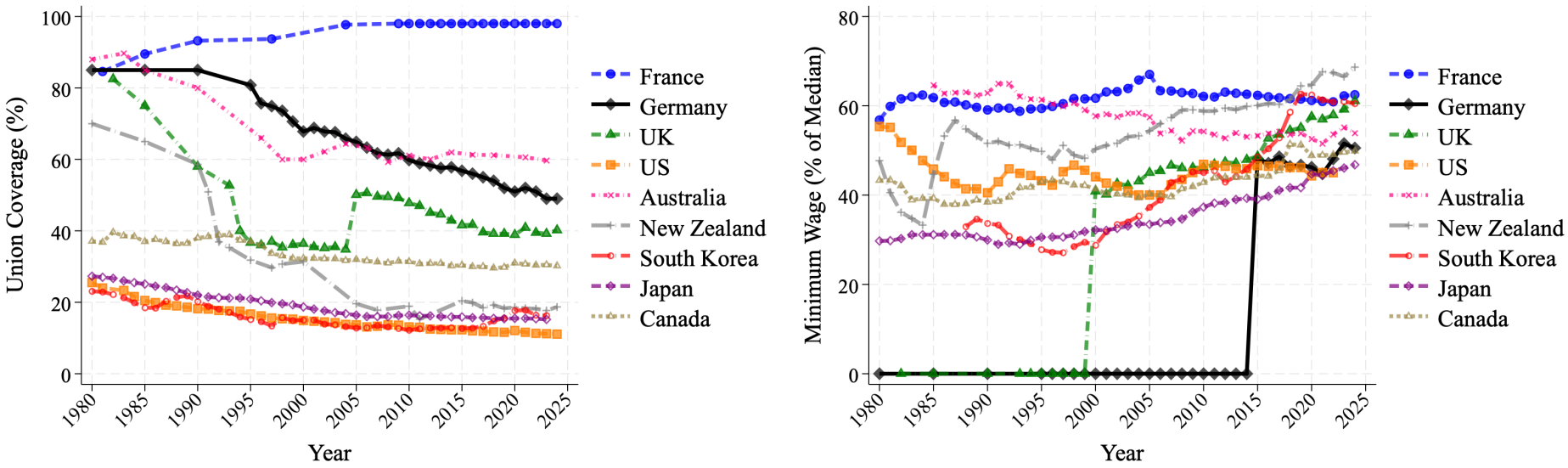
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- But how are these two trends related?

Minimum Wage and Collective Bargaining Coverage Trends



(a) Collective Bargaining Coverage

(b) Minimum Wage as % of Median

Source: OECD and authors' own calculations

- Simultaneous decline in collective bargaining coverage and increased reliance on minimum wages over time
- But how are these two trends related?
- Germany is a great case study - recent minimum wage introduction and a strong history of collective bargaining.

Minimum Wage and Coverage Interactions

- Historically unions have strong and evolving views on minimum wages:
  - From resistance and fears of "Tarifflucht": *"Collective bargaining is a matter for the social partners, not the state..We are not in favor of legally mandated minimum wages ."* (Hubertus Schmoldt, IG BCE, 2005) [1]

[1] https://www.focus.de/politik/deutschland/einfach-zu-billig-mindestlohn_id_2030289.html
[2] https://igbau.de/Binaries/Binary6418/pm-55-2014-Mindestlohn-verabschiedet-3.7.2014.pdf
[3] https://www.theguardian.com/world/2020/jan/12/nordic-countries-at-odds-with-eu-over-minimum-wage

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  - To support: *"The vote [to introduce the minimum wage] shows the will of the MPs to turn the conditions on the labour market from their head back onto their feet."* (Robert Feiger, IG BAU, 2014) [2]

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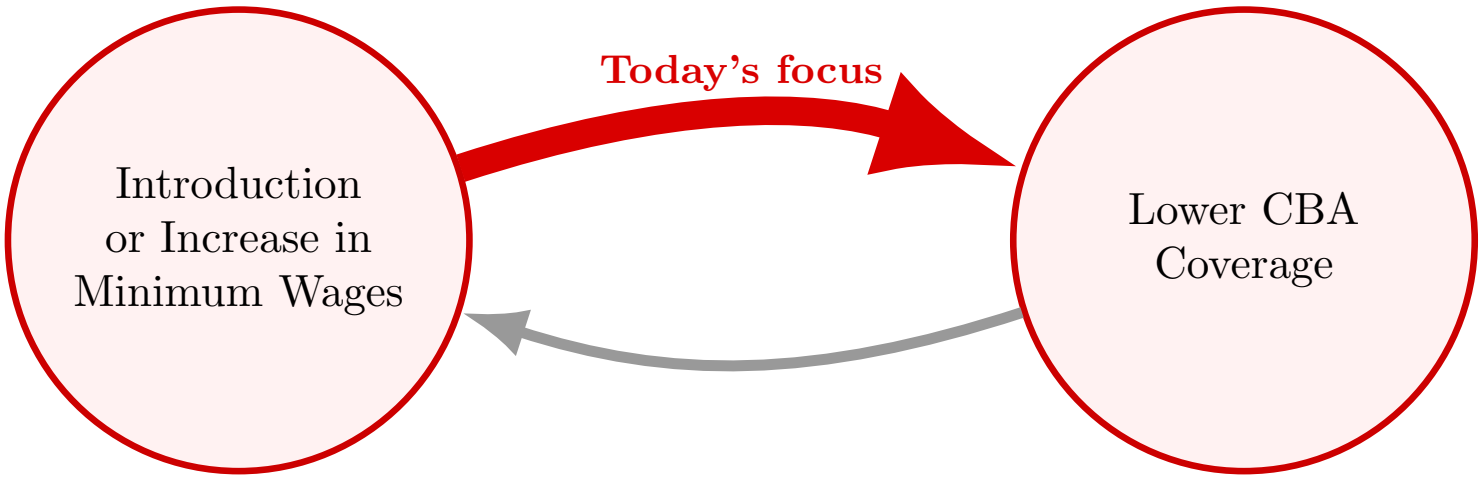
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- These interactions are a live policy issue in the EU:
  - *"The Danish Confederation of Trade Unions .. fears that a statutory minimum wage can weaken the effectiveness of the Danish model."* (Lizette Risgaard, president of the Danish trade union confederation, 2020) [3]

[1] https://www.focus.de/politik/deutschland/einfach-zu-billig-mindestlohn_id_2030289.html
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Conceptual Link: Minimum Wage (MW) and Coverage



- Two research q's related to German minimum wage (MW) introduction:
  1. What were the impacts on coverage?
  2. What mechanisms explain these impacts?
     - MW impacts on collective bargaining negotiations (CBNs), *exploiting newly digitized CBN data on union demands and outcomes*
     - Role of monopsony in mediating impacts

Overview

- **RQ1:** Causal impact of minimum wages on coverage
  - Method: Firm and worker coverage using event-study DiD design. Treatment based on pre-reform sector-region exposure to MW
  - *Finding: MW intro. leads to a 4.8% point lower coverage for more exposed establishments by 2019 (4.6% for workers), vs. baseline coverage of 29.7% for establishments (56.7% for workers).*

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  - Method: Estimate heterogeneous MW effects on coverage by employment HHI
  - *Finding: Negative coverage impacts concentrated in top HHI tercile.*

Related Literature

- **Minimum Wage Impacts**
  - Large minimum wage literature estimates wage/employment effects (e.g. Dustmann et al. (2021); Cengiz et al. (2019))
  - Monopsony literature shows MW can increase employment and wages (e.g. Popp (2024); Azar et al. (2024)). Also see Faia et al. (2026) for MW impacts on markdowns
  - *Contribution: show MW impacts on coverage, and impact increases with monopsony power (concentration).*
- **Minimum Wage and Collective Bargaining Interactions**
  - US - Clemens and Strain (2025)
  - Germany - Bellmann et al. (2021); Bispinck et al. (2020); Fulda et al. (2023)
  - *Contribution: show early announcement impacts of MW on coverage, and impacts on CBNs as a mechanism explaining the negative coverage effect.*

Outline

1. Motivation
2. Institutional Background
3. The Coverage Decision
4. Data
5. Empirical Strategy
6. Results

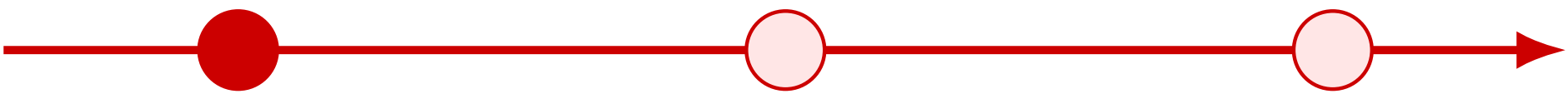
Institutional Background

Minimum Wage Policy in Germany

2013
Election and coalition agreement including commitment to a minimum wage

2014
Legislation passed introducing the minimum wage

2015
Introduction of the minimum wage (€8.50/hour)



The Bargaining Process

- Unions and employer associations negotiate CBAs at the sector-by-region level (more rarely at firm level)
- Wages are negotiated annually or biannually
- Framework agreements have a longer term and cover non-wage amenities: training, holidays, hours and job separation terms
- *Firms decide whether to sign on to CBAs through their employer associations - not mandatory*
- Once signed on to a CBA, terms apply for duration even if firm exits

The Coverage Decision

Firm Supply of Coverage

- Decision to sign on to CBA ultimately rests with firm
- Coverage may lower admin costs of setting employment terms, and provide dispute resolution mechanisms (Jäger, Noy and Schoefer (2022))
- Given pay and amenity premiums of opting into coverage, coverage decision similar to deciding to post high rewards package
- Benefits: higher labour supply (recruitment, retention or effort). Costs: more expensive per worker hired
- This intuition holds both in static and dynamic monopsony models ((Burdett and Mortensen, 1998; Manning, 2006; Card et al., 2018; Caldwell et al., 2026))
- Typically these models assume firms can perfectly commit to terms posted to workers - joining a CBA is one way to achieve this commitment

Firm Supply of Coverage and the Minimum Wage

- Monopsony models suggest lower wage offers/rewards (higher markdowns) when markets are less competitive and labour supply less elastic
- Extrapolating from such theories gives us three hypotheses (yet to be grounded in formal model):
- *(H1) We would expect lower coverage in more monopsonistic markets, all else equal, as less competitive pressures to offer coverage*
- (H2) MW intro will lower coverage as it either lowers benefits of coverage - by lowering coverage recruitment premium - or increases costs of coverage if MW spillovers to CBA wages
- (H3) Any reductions in coverage following MW intro are likely to be concentrated in more monopsonistic markets where firms face less competitive pressure to maintain coverage

Worker Demand

Data

Data

- **IAB Establishment Panel:**
  - Annual survey of ∼15k firms
  - **Key variables:** CBA coverage, count of minimum wage employees, count of employees
- **Admin. Worker Data Merged Establishment Panel (LIAB QM)**:
  - **Key variable:** Daily wages at individual worker level, sector of employment. Coverage status of employees
- **Collective Bargaining Negotiation (CBN) Data:**
  - We digitized data on trade-union demands and outcomes from CBNs, obtained from WSI (Gergs, Kügler, and Zentler-Munro, 2026)
  - We use web-scraped data on detailed CBA wage floors from *Tarifdatenbank* of the German Statistical Office (Destatis)
  - **Key variables:** Wage floors (Destatis), wage growth demands and outcomes (WSI)

Empirical Strategy

Empirical Strategy

- Differences-in-differences design using variation in exposure to the new min. wage policy measured in 2014

- **Exposure Measure:**
  - Share of workers affected by the Minimum Wage (as stated by establishments) in 2014, at sector-by-state level

$$\text{Exp}_{s,r} = \frac{\sum_{j \in s,r} Employees_j^{MW}}{\sum_{j \in s,r} Employees_j}$$

  - $Employees_j^{MW}$ is number of employees at establishment j with wage $<$ €8.50 (as reported by estab in 2014 BP), $Employees_j$ is total number of employees, s is sector, r is state

Empirical strategy

- **Event Study Specification:**
  - Treatment group: whether establishment (or worker) is in sector-by-state cells in the top three quartiles of exposure distribution $T_{sr} = 1$
  - Control group: Sector-by-state cells in the first quartile $T_{sr} = 0$

$$y_{isrt} = \alpha_{sr} + \gamma_t + \left(\sum_{k \neq 2012} \beta_k \cdot 1\{t = k\} \cdot T_{sr} \right) + \epsilon_{isrt}$$

  where i is establishment (worker), sr is sector-state of i, t is year, establishment regressions weighted by number of employees
  - Main outcome of interest y_{isrt} is CBA coverage

- **Quantile Pre/Post Specification:**

$$y_{isrt} = \gamma_t + \sum_{q \neq 1} \beta_q \cdot 1\{i \in q\} + \sum_{q \neq 1} \beta_q^{post} \cdot 1\{i \in q\} \cdot Post_{2012} + \epsilon_{isrt}$$

- q is quartile for most analyses

Descriptive statistics

	Treatment	Control
N (firms)	8,474	5,074
N (individuals)	775,097	816,744
2012 characteristics		
CBA coverage (firms)	0.297	0.364
	(0.457)	(0.481)
CBA coverage (workers)	0.567	0.640
	(0.495)	(0.480)
Log employment	1.982	1.986
	(1.061)	(1.137)
Log wages	2.185	2.569
	(0.967)	(0.880)
Manufacturing	0.064	0.161
	(0.245)	(0.368)
Services	0.935	0.831
	(0.246)	(0.374)

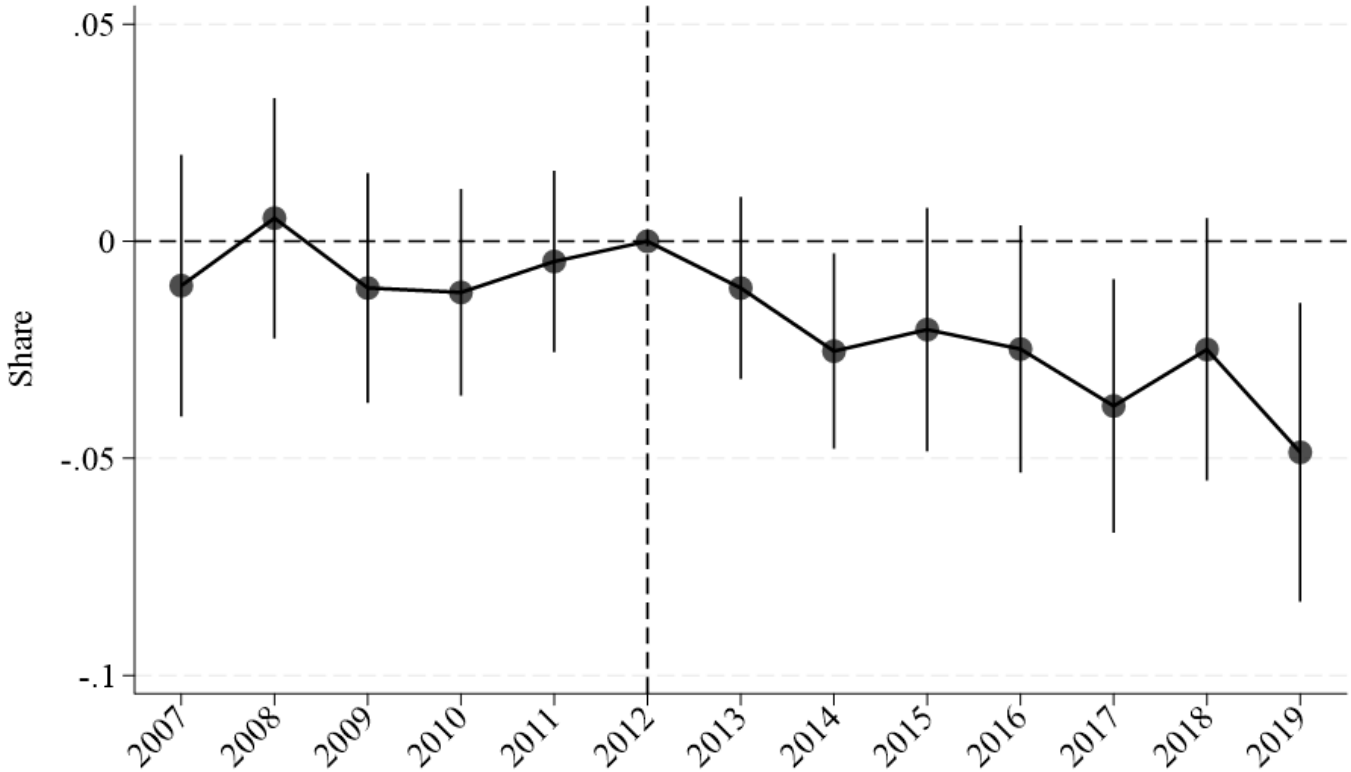
Notes: Standard deviations in parentheses. All means and standard deviations are survey-weighted. Treatment cells are sector × state cells in the top three quartiles of minimum-wage exposure; control cells are in the bottom quartile.

Results

Road Map for Results

- **Direct Result:**
  - Minimum Wage Announcement → Lower Coverage
- **Mechanism Result:**
  - Minimum Wage Announcement → CBN Negotiations → Lower Coverage
- **Mediation Result:**
  - Minimum Wage Announcement → Pre-MW Concentration→ Lower Coverage
  - Using employment HHI as proxy for competitiveness

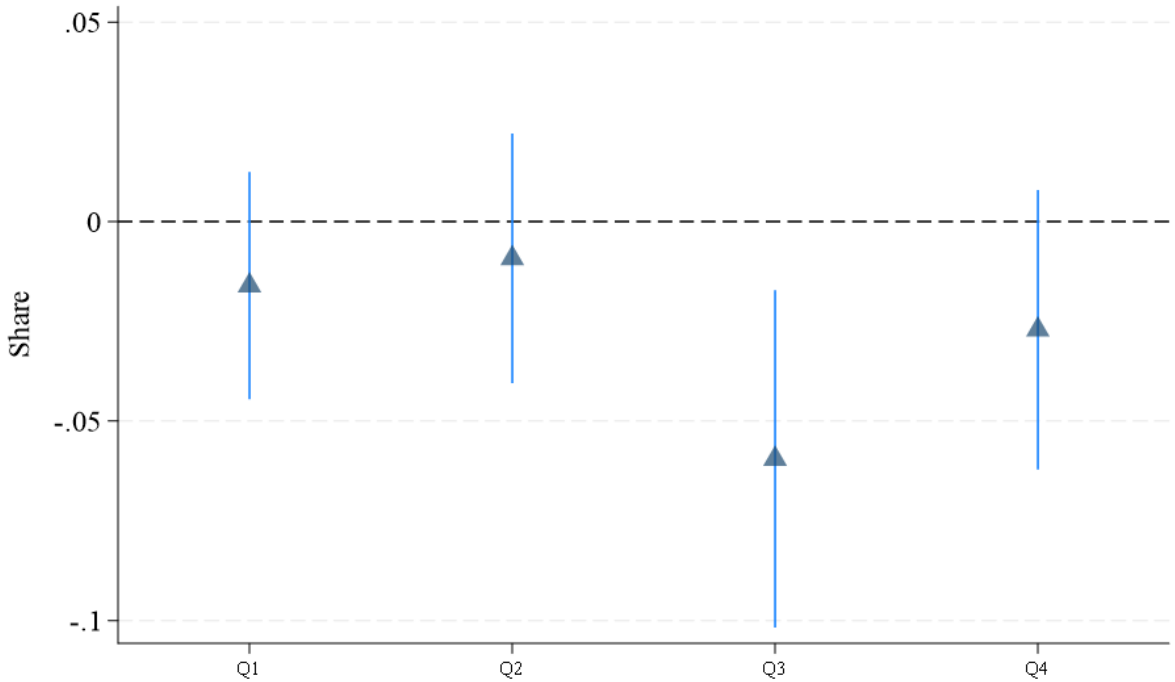
Direct Results: Establishment Coverage



$$Cov_{isrt} = \alpha_{sr} + \gamma_t + \left(\sum_{k \neq 2012} \beta_k \cdot 1\{t = k\} \cdot T_{sr} \right) + \epsilon_{isrt}$$

- Coverage decreases by 5% pts by 2019, vs baseline of 30% Opt-In vs Opt-Out

Direct Results: Coverage Effect Heterogeneity by Exposure Quartile



$$Cov_{isrt} = \gamma_t + \sum_{q \neq 0} \beta_q \cdot 1\{i \in q\} + \sum_{q \neq 0} \beta_q^{post} \cdot 1\{i \in q\} \cdot Post_{2012} + \epsilon_{isrt}$$

- Here quartiles are among treated sector-state cells, with Q1 as the least exposed and Q4 as the most exposed. Cells with zero exposure ($q = 0$) are the reference category
- Coverage decrease is largest in the top half of exposure distribution

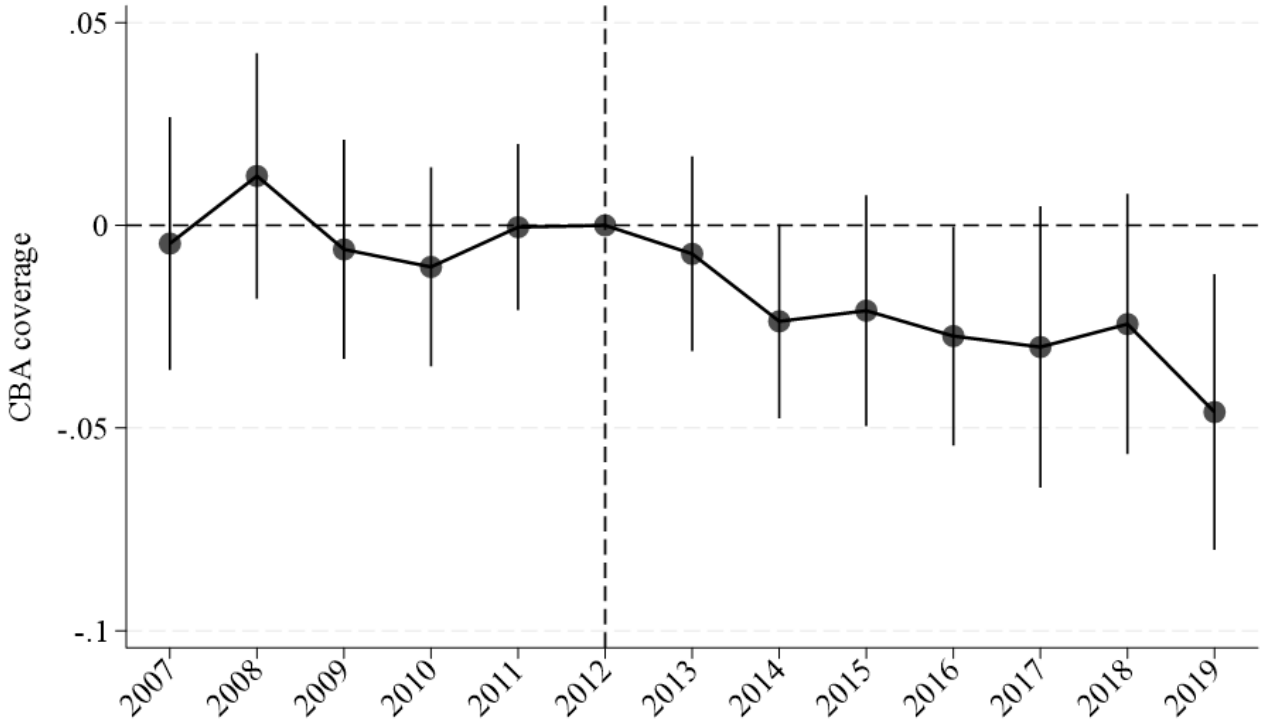
Direct Results: Robustness of Coverage Effect

	(1)	(2)	(3)	(4)
Post2012 × Treat	−0.0223*	−0.0202*	−0.0279*	
	(0.0100)	(0.0100)	(0.0114)	
Post2014 × Treat				−0.0225*
				(0.0101)
Sector × state FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Control for estab. foreign-worker share	No	Yes	No	No
Extension sectors excluded	No	No	Yes	No
Treatment year	2013	2013	2013	2015
N	170,877	170,877	124,863	170,877

*$p < 0.05$.

- Baseline results (Col 1) robust to:
  - Col 2: Controlling for establishment-level share of foreign workers, addressing potential confounding from opening of Germany to EU-8 accession states in 2011
  - Col 3: Excluding extension sectors, where CBA is extended to all establishments, regardless of membership of CBA
  - Col 4: Using a post-event indicator of 2015 and onwards

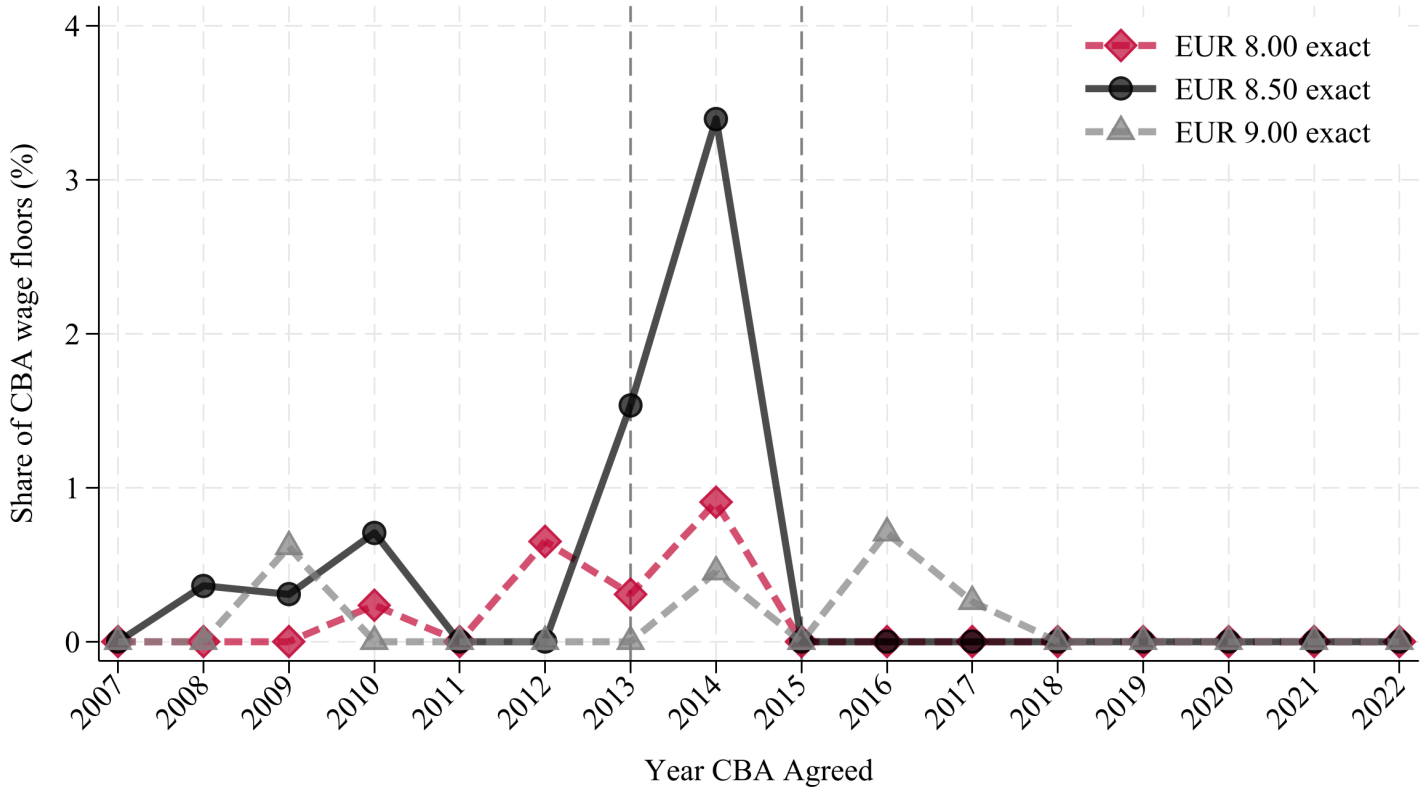
Direct Results: Worker Coverage



$$Cov_{isrt} = \alpha_{sr} + \gamma_t + \left(\sum_{k \neq 2012} \beta_k \cdot 1\{t = k\} \cdot T_{sr} \right) + \epsilon_{isrt}$$

- Similar size decrease in worker coverage to firm coverage
- Suggests workers not reallocating to covered firms but instead losing coverage within firms

Mechanism Results: Spillovers via CBNs, Announcement Effects



Dashed lines: 2013 announcement and 2015 statutory introduction.

- CBA wage floors show increased bunching at the €8.50 threshold as early as 2013
- Negotiated outcomes respond quickly to policy communication as well as legislation

Mechanism Results: Wage Floors

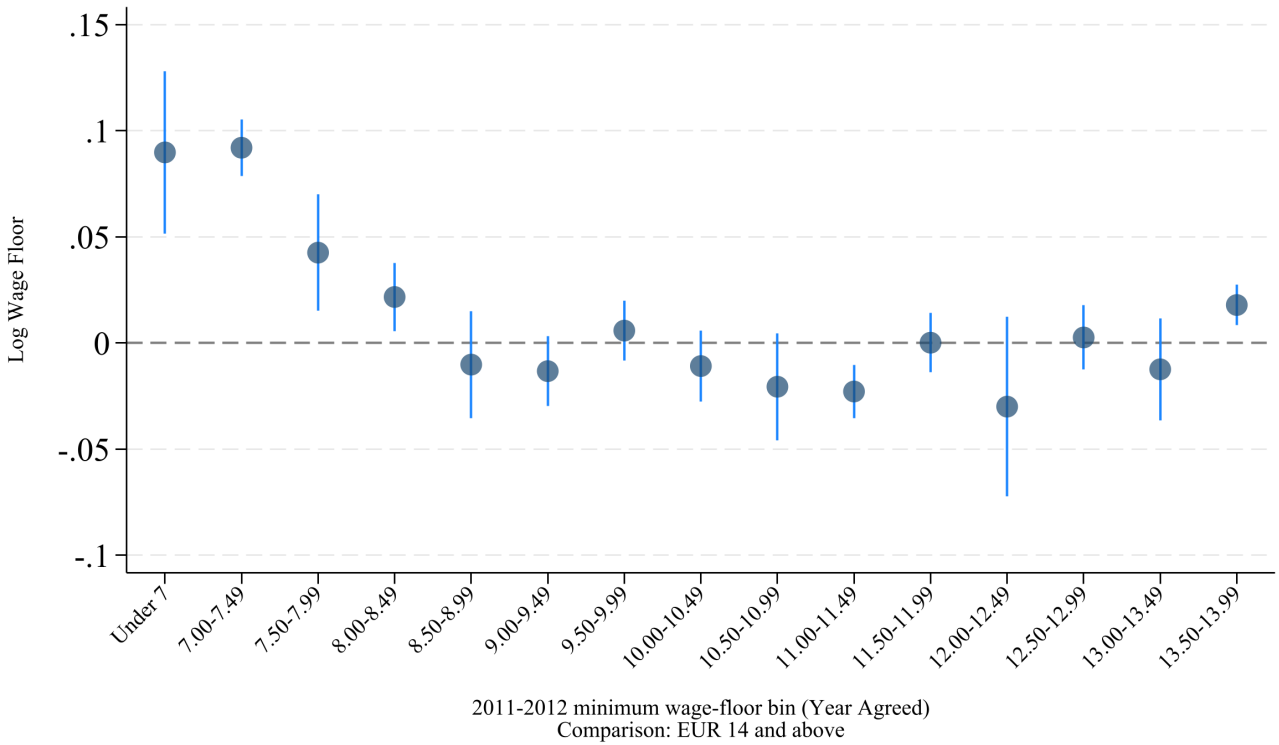
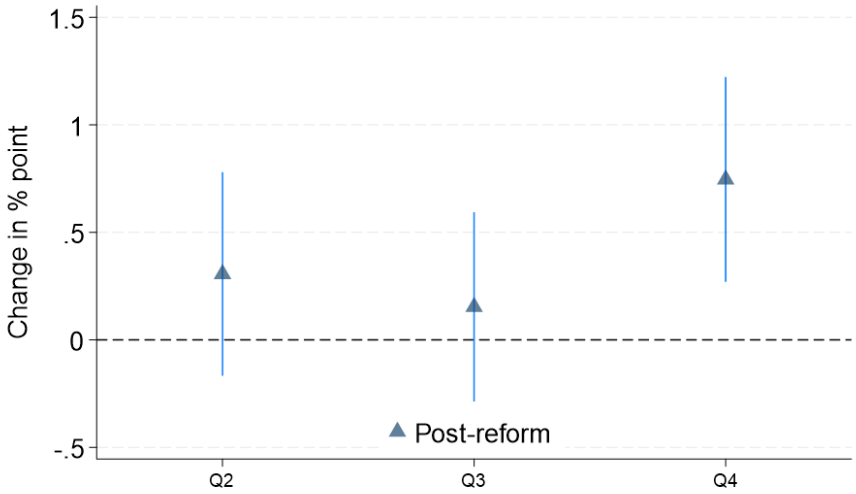


Figure: Year-agreed log real wage floors by 2011-2012 wage-floor bin

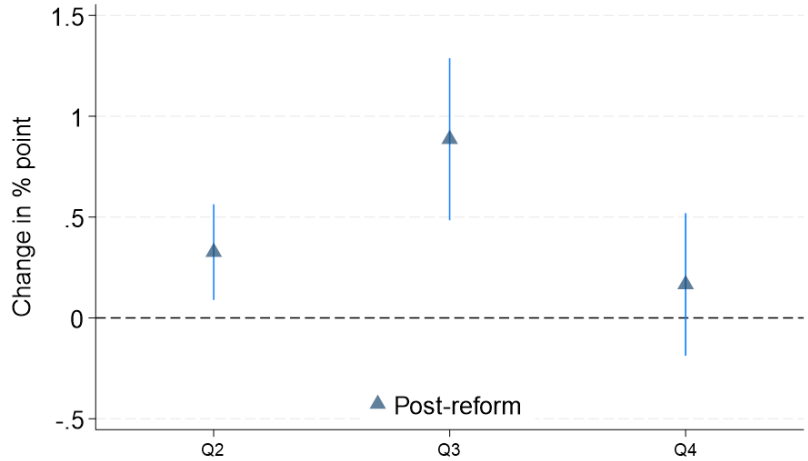
$$Y_{it} = \alpha_i + \lambda_t + \sum_{b \in \mathcal{B}} \beta_b \left(1\{WageFloor_{i,2011-12} = b\} \times 1\{t \geqslant 2013\}\right) + \varepsilon_{it}$$

Event Study

Mechanism Results: Spillovers via Wage Growth in CBNs



Demands

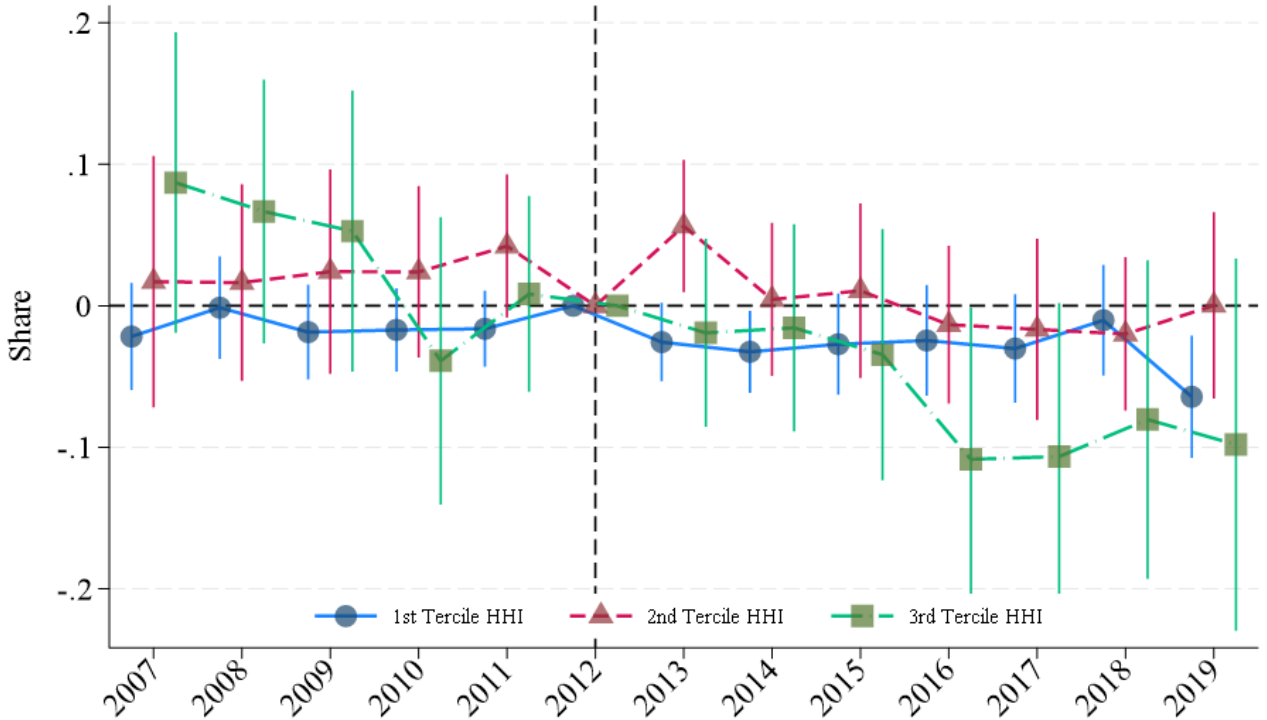


Outcomes

$$y_{isrt} = \gamma_t + \sum_{q \neq 1} \beta_q \cdot 1\{i \in q\} + \sum_{q \neq 1} \beta_q^{post} \cdot 1\{i \in q\} \cdot Post_{2012} + \epsilon_{isrt}$$

- y_{isrt} is % wage growth demand (outcome) in CBNs, applicable to those at *and above* wage floors
- Demands: MW impact peaks for Q4 - 0.75% pts increase vs 6% baseline
- Outcomes: MW impact peaks for Q3 - 0.9% pts increase vs 3.2% baseline
- Suggests CBNs lead to spillover effects of MW - could incentivize firms to exit coverage

Mediator Results: Monopsony, MW and Coverage



$$Cov_{isrt} = \alpha_{sr} + \gamma_t + \left(\sum_{k \neq 2012} \beta_k \cdot 1\{t = k\} \cdot T_{sr} \right) + \epsilon_{isrt}$$

- Negative coverage impacts located in most concentrated markets (HHI based on employment stocks)

Mediator Results: Monopsony, MW and Coverage

Table: DiD Estimates: Coverage Effects

	DiD	DiD + HHI
Post2012 × Treat	-0.0223*	-0.0184
	(0.0100)	(0.0118)
Post2012 × Treat × HHI tercile 2		0.0008
		(0.0317)
Post2012 × Treat × HHI tercile 3		-0.0706*
		(0.0353)
Sector × State FE	Yes	Yes
Year FE	Yes	Yes
Baseline cov. (treated, 2012)	0.297	0.297
N	170,877	170,688

Notes: Standard errors in parentheses. *$p < 0.05$.

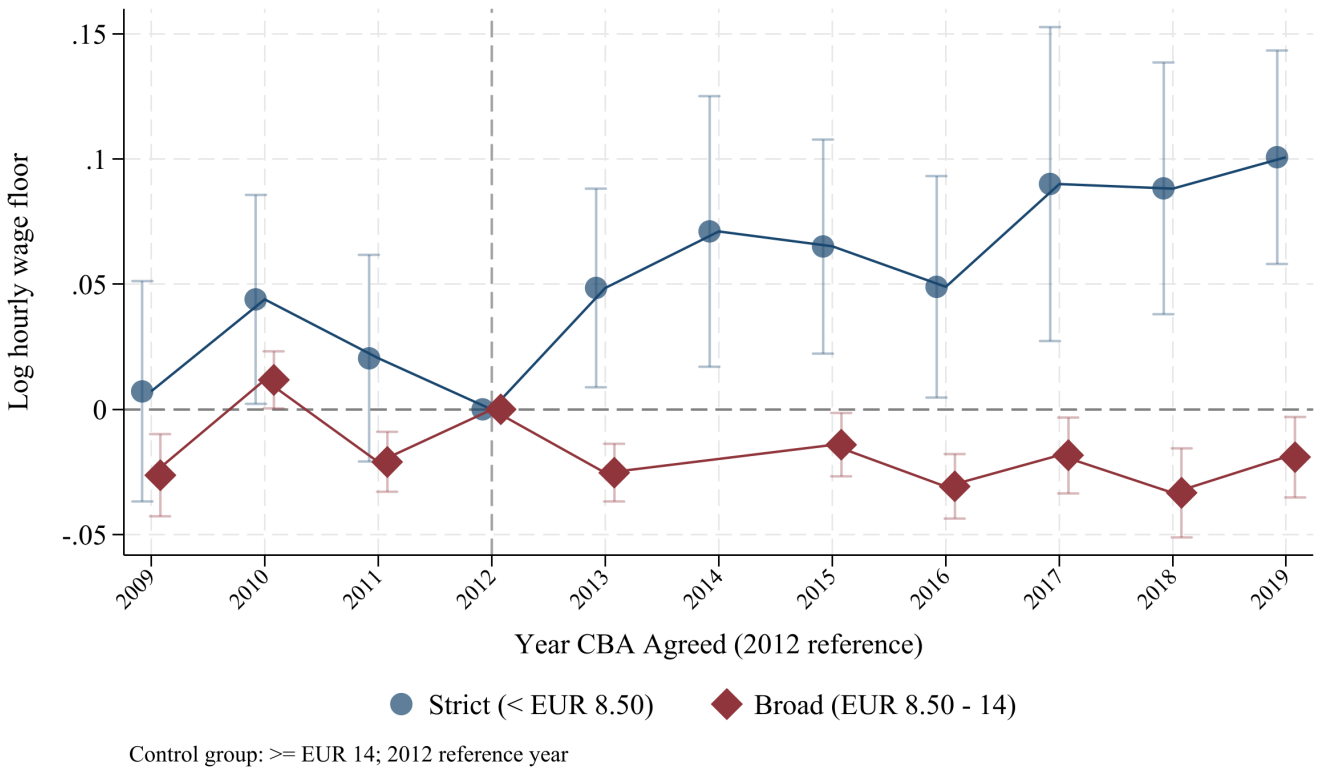
Conclusion

- **Key Result: Minimum Wage introduction reduced collective bargaining coverage**
  - Establishments in more exposed sectors/regions saw coverage fall by 5 percentage points by 2019, around 15% of baseline coverage
- **Key Mechanism: Minimum Wage changed collective bargaining dynamics**
  - Led to higher wage growth demands and outcomes in CB negotiations, with significant spillover effects
- **Key Mediator: reductions in coverage greatest in more concentrated markets**
  - Negative coverage effects are greatest in most concentrated markets (top HHI tercile)

Appendix

Back to Results

Appendix: Event Study of Log Wage Floor by Year Agreed



$$Y_{it} = \alpha_i + \lambda_t + \sum_{k \in \mathcal{K}} \left[\beta_k^S 1\{t = k\} Strict_i + \beta_k^B 1\{t = k\} Broad_i \right] + \varepsilon_{it}, \quad 2012 \notin \mathcal{K}$$

Y_{it} is the log real wage floor; $Strict_i$ denotes a 2011-2012 minimum below €8.50, $Broad_i$ denotes €8.50 to below €14, and the omitted control group is €14 and above. Agreement-region and year fixed effects; weighted by states covered and clustered by agreement-region.

Back to DiD

Why do firms sign on to CBAs?

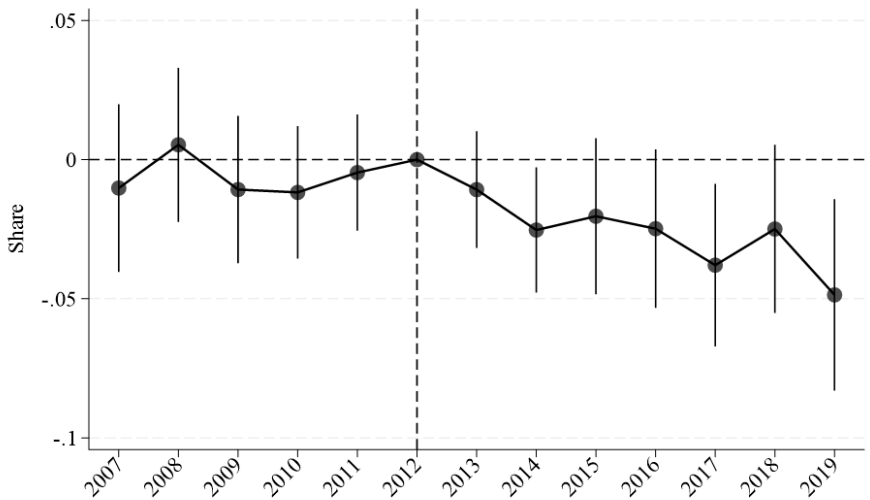
Jäger, Noy and Schoefer (2022) note three potential reasons:

1. Worker pressure

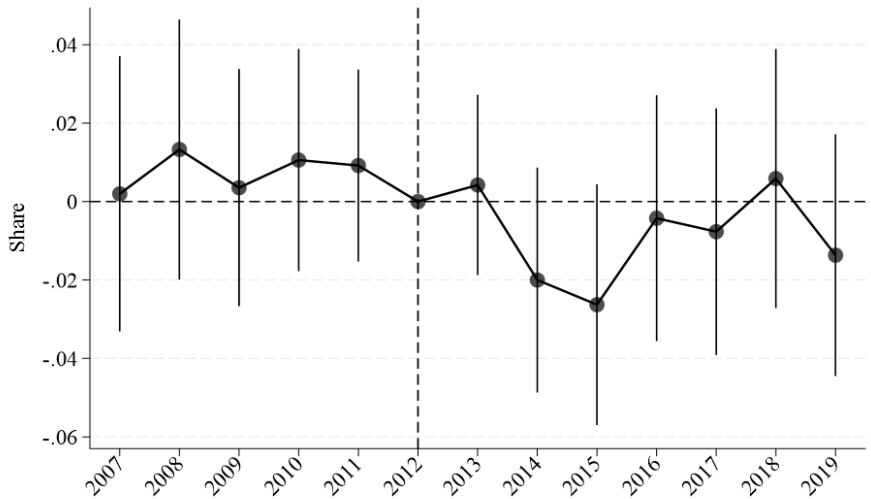
2. Other benefits from the employer association (e.g. legal advice, lobbying, etc.)
   - But firms can join employer associations without signing on to CBAs

3. Peaceful dispute resolution

Appendix: Establishment Coverage and Heterogeneity by Size



(a) Coverage effect, employment-survey weighted

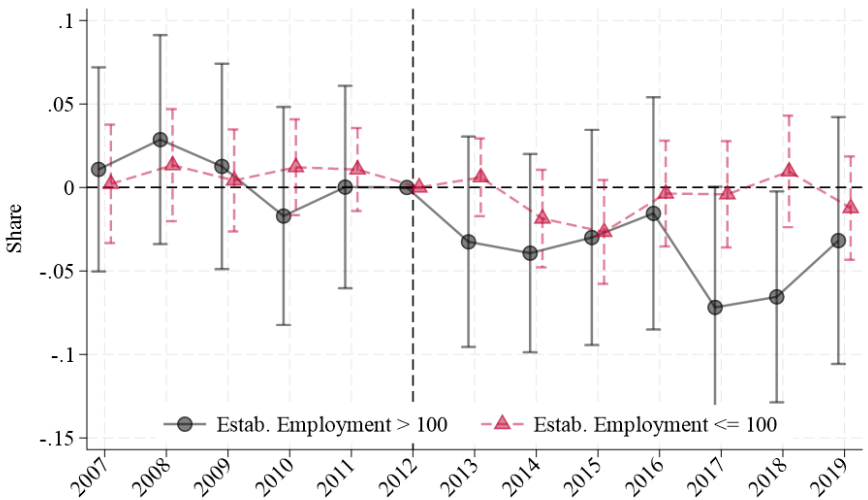


(b) Coverage effect, survey weighted

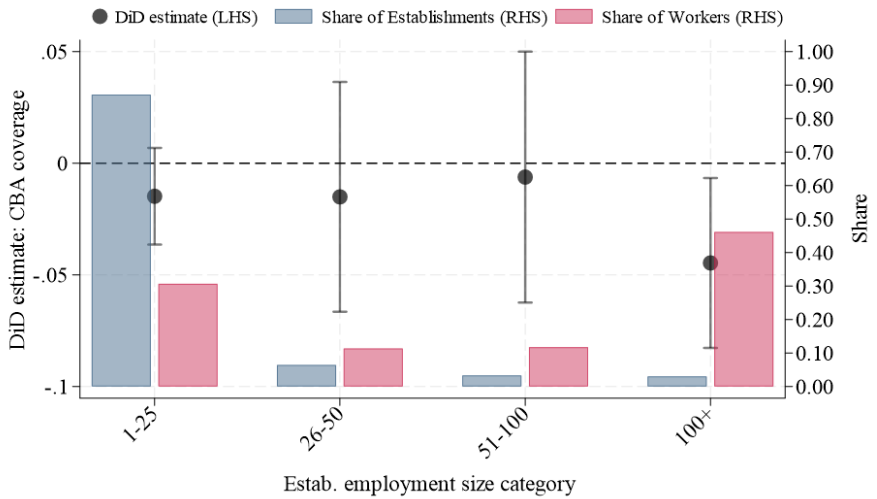
Figure: Minimum-wage effects on CBA coverage.

$$Cov_{isrt} = \alpha_{sr} + \gamma_t + \left(\sum_{k \neq 2012} \beta_k \cdot 1\{t = k\} \cdot T_{sr} \right) + \epsilon_{isrt}$$

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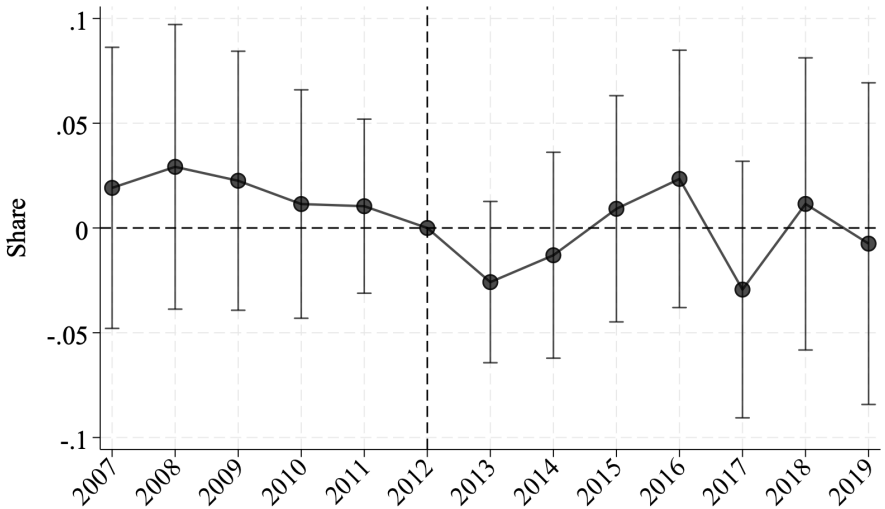
(a) Large versus small firms



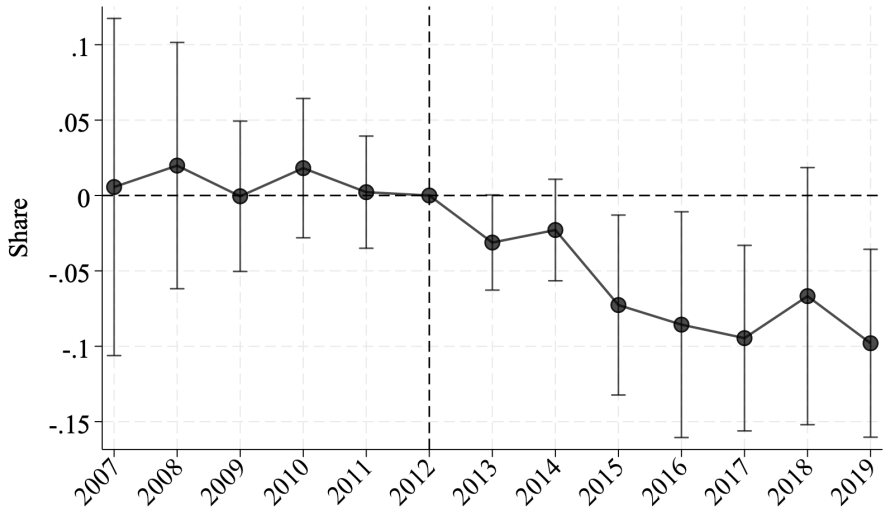
(b) Coverage response by employment-size category

Figure: Size profile of the coverage response.

Appendix: Coverage Effect Heterogeneity by Sector



Manufacturing

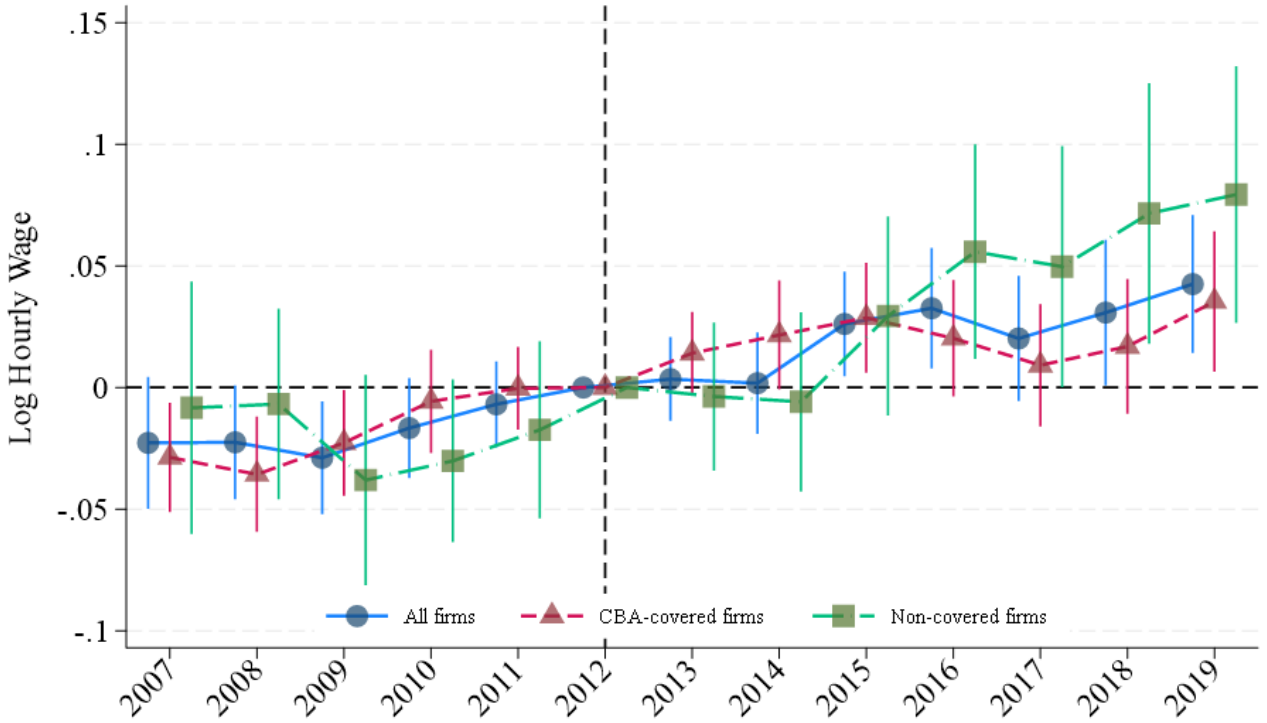


Services

$$Cov_{isrt} = \alpha_{sr} + \gamma_t + \left(\sum_{k \neq 2012} \beta_k \cdot 1\{t = k\} \cdot T_{sr} \right) + \epsilon_{isrt}$$

- Coverage decrease most pronounced in services sector

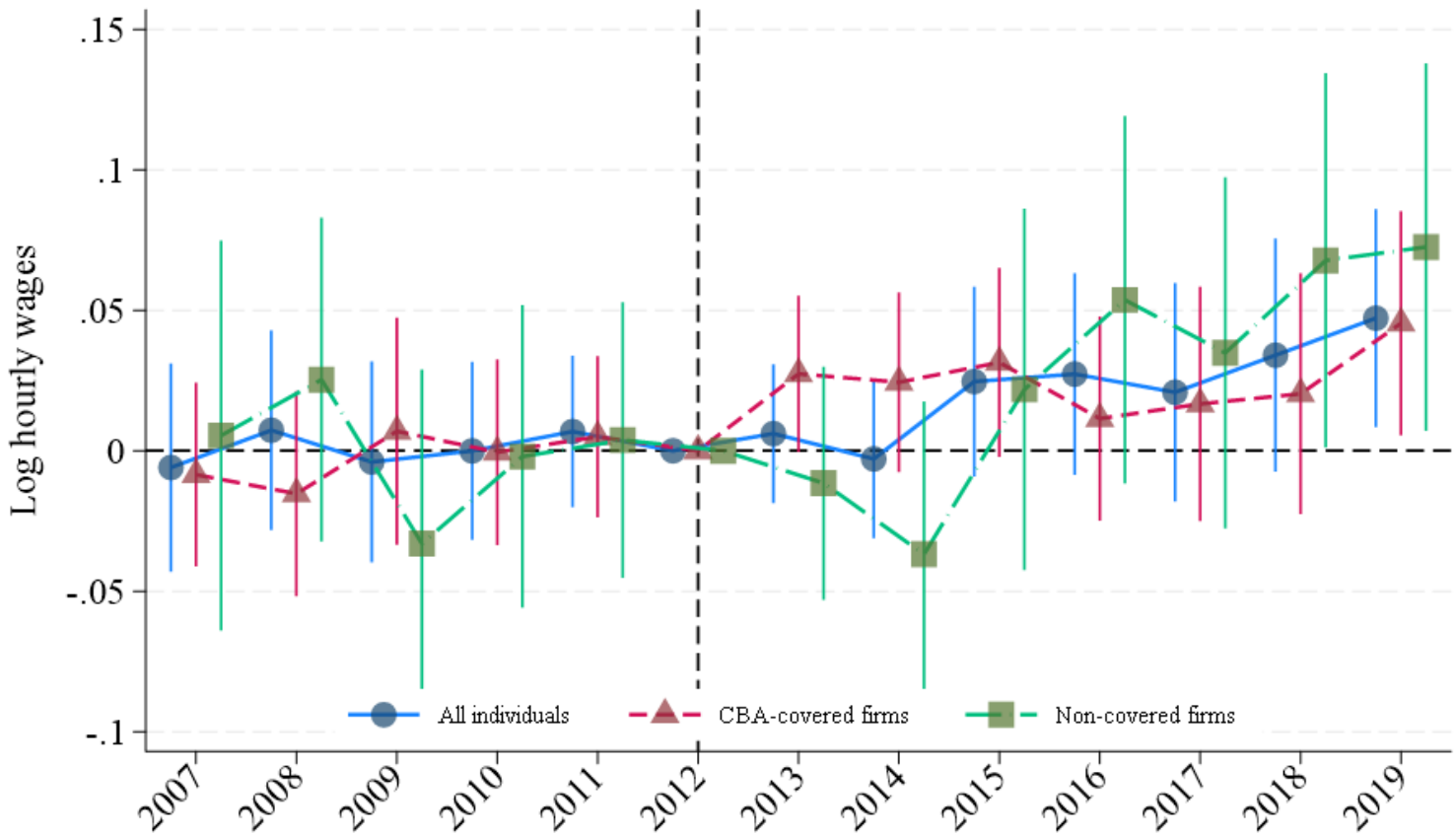
Appendix: Wage Impacts (Establishments)



$$y_{isrt} = \alpha_{sr} + \gamma_t + \left(\sum_{k \neq 2012} \beta_k \cdot 1\{t = k\} \cdot T_{sr} \right) + \epsilon_{isrt}$$

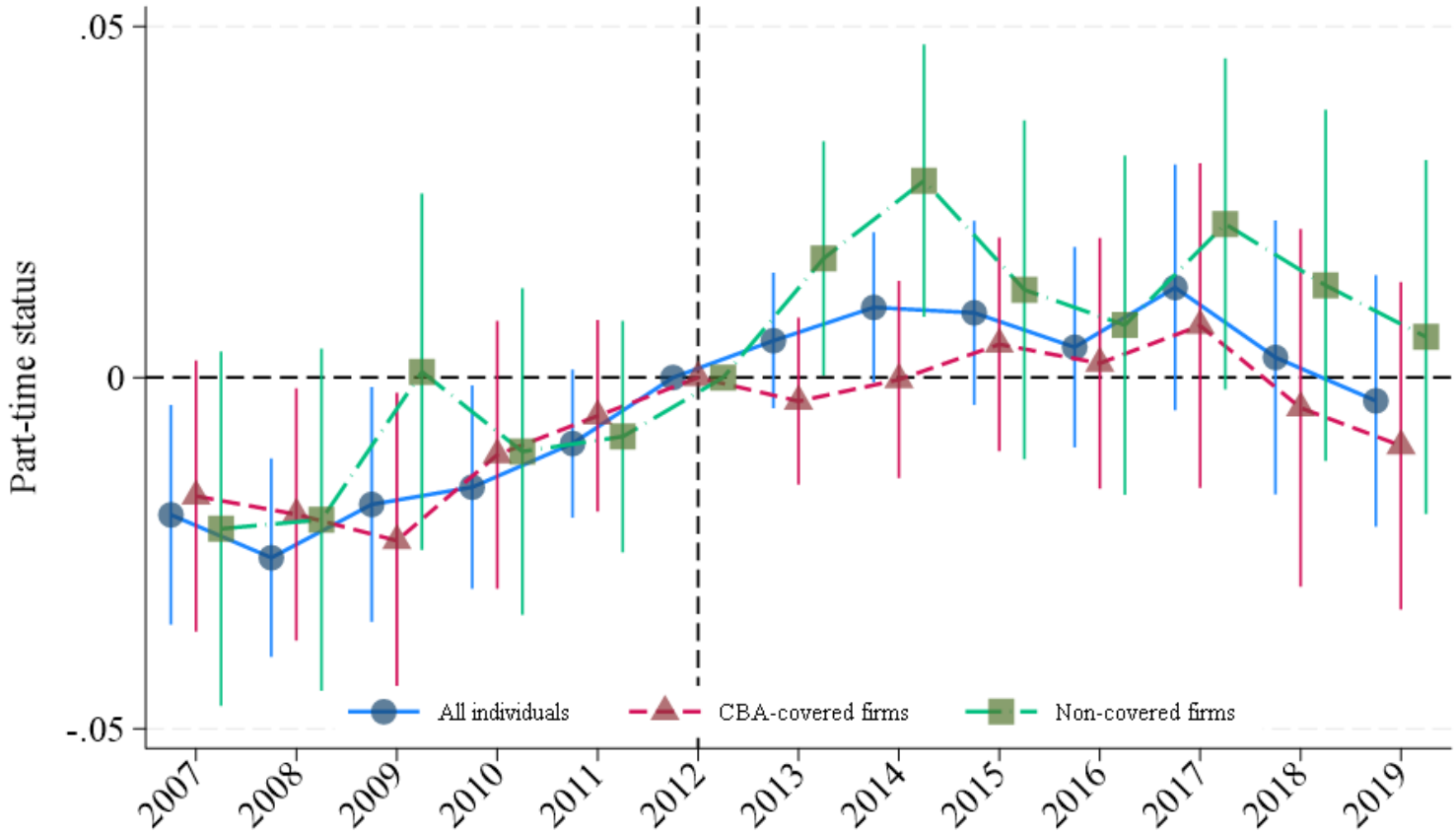
- MW impacts on wages bigger for non-covered establishments and workers
- Wages increase later than coverage decrease
- Dynamics in line with profit maximisation Workers Wages

Appendix: Wage Impacts (Workers)



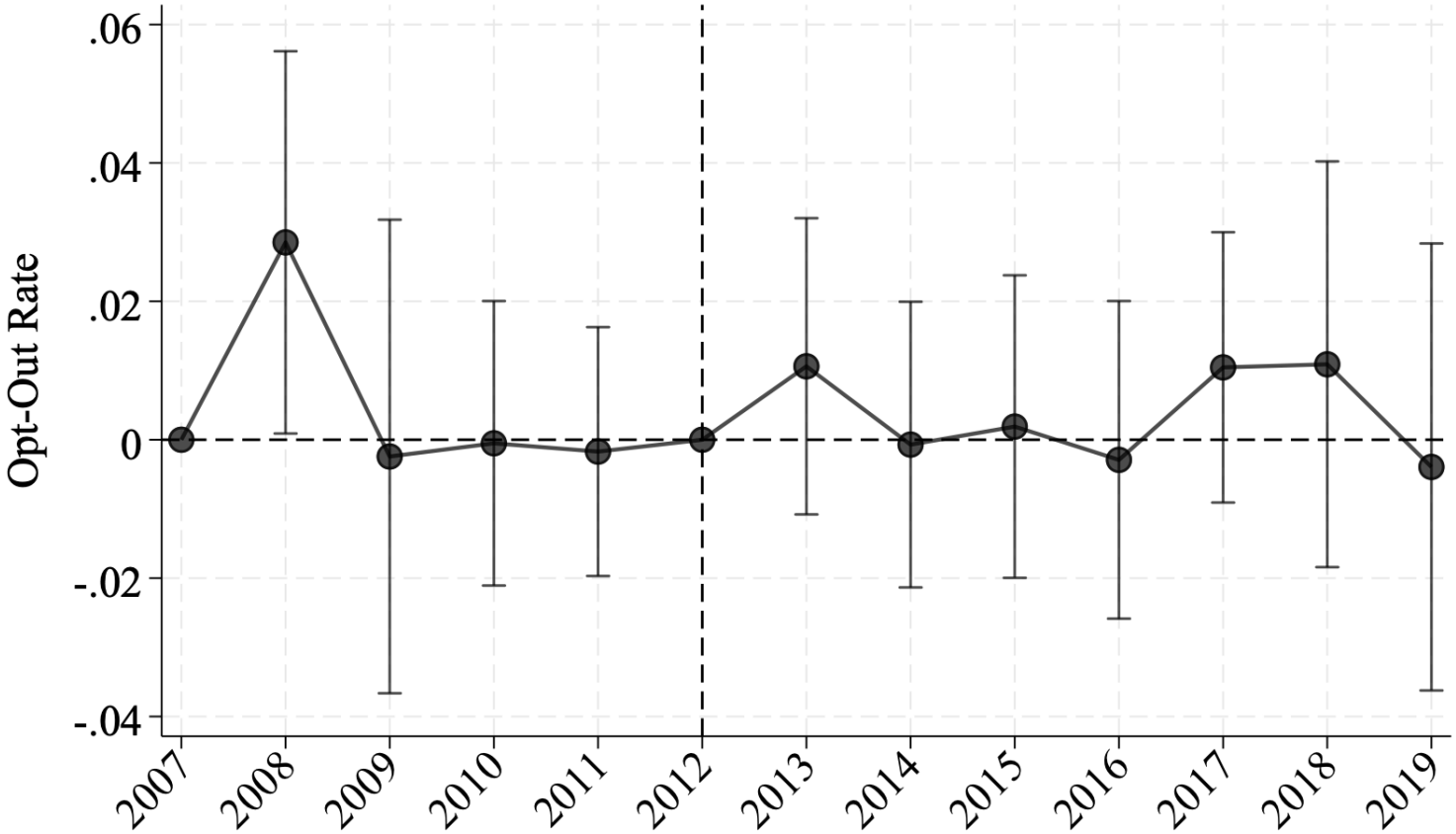
$$y_{isrt} = \alpha_{sr} + \gamma_t + \left(\sum_{k \neq 2012} \beta_k \cdot 1\{t = k\} \cdot T_{sr} \right) + \epsilon_{isrt}$$

Appendix: Part-Time Employment (Workers)



$$y_{isrt} = \alpha_{sr} + \gamma_t + \left(\sum_{k \neq 2012} \beta_k \cdot 1\{t = k\} \cdot T_{sr} \right) + \epsilon_{isrt}$$

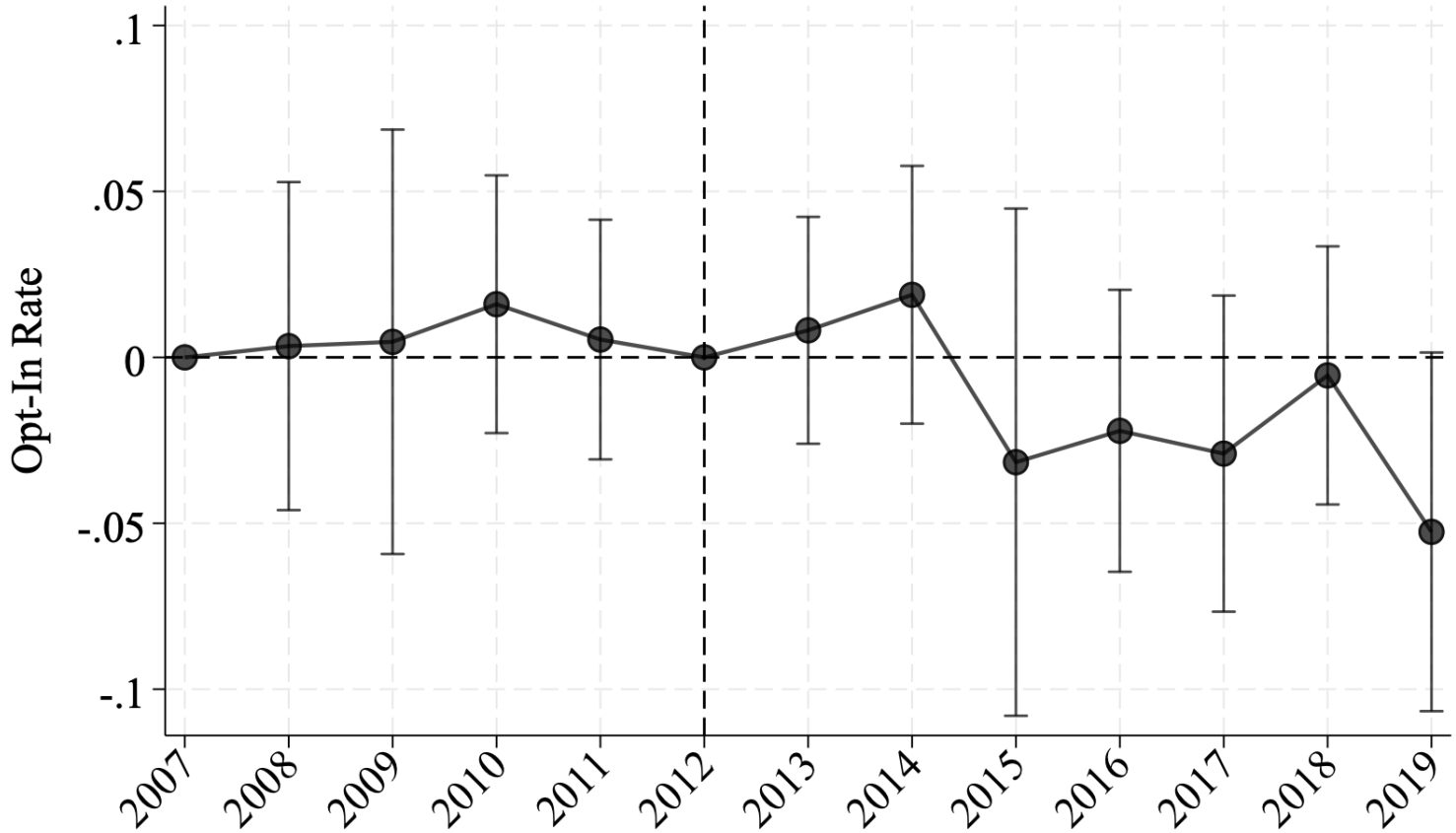
Appendix: Opt-Out, Event Study



$$y_{isrt} = \alpha_{sr} + \gamma_t + \left(\sum_{k \neq 2012} \beta_k \cdot 1\{t = k\} \cdot T_{sr} \right) + \epsilon_{isrt}$$

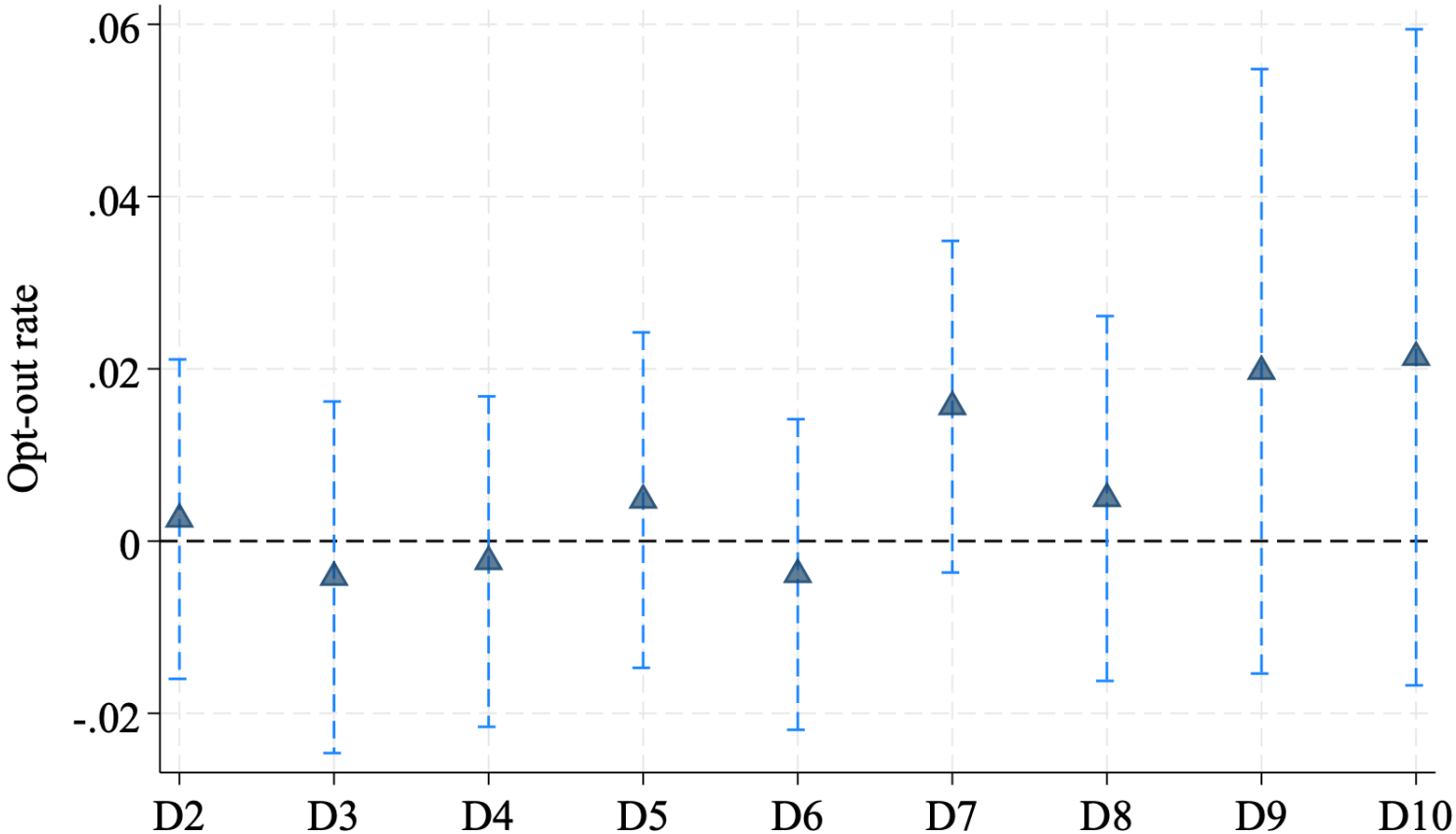
Back to Results

Appendix: Opt-In Event Study



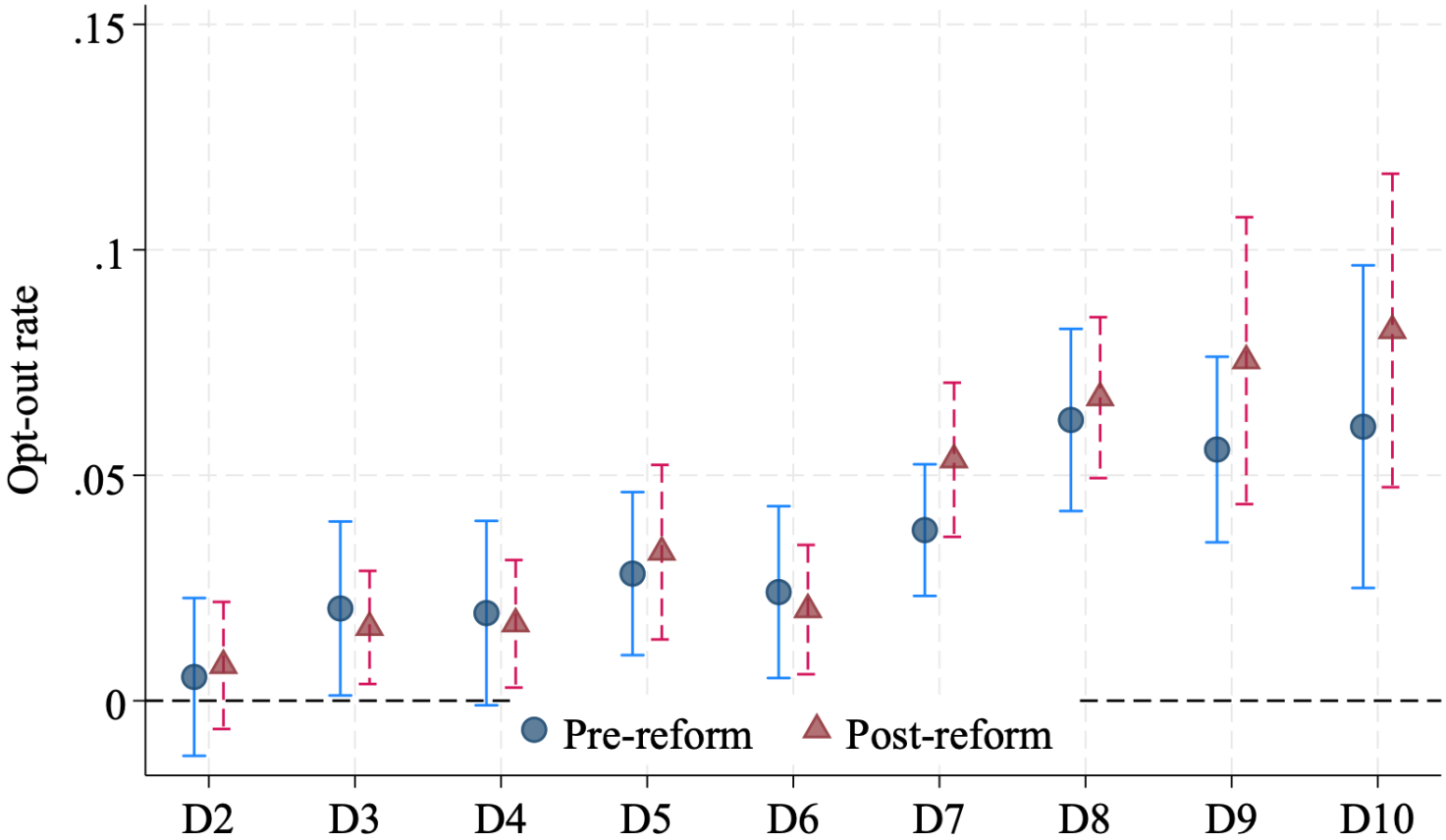
$$y_{isrt} = \alpha_{sr} + \gamma_t + \left(\sum_{k \neq 2012} \beta_k \cdot 1\{t = k\} \cdot T_{sr} \right) + \epsilon_{isrt}$$

Appendix: Opt-Out, by Exposure Decile



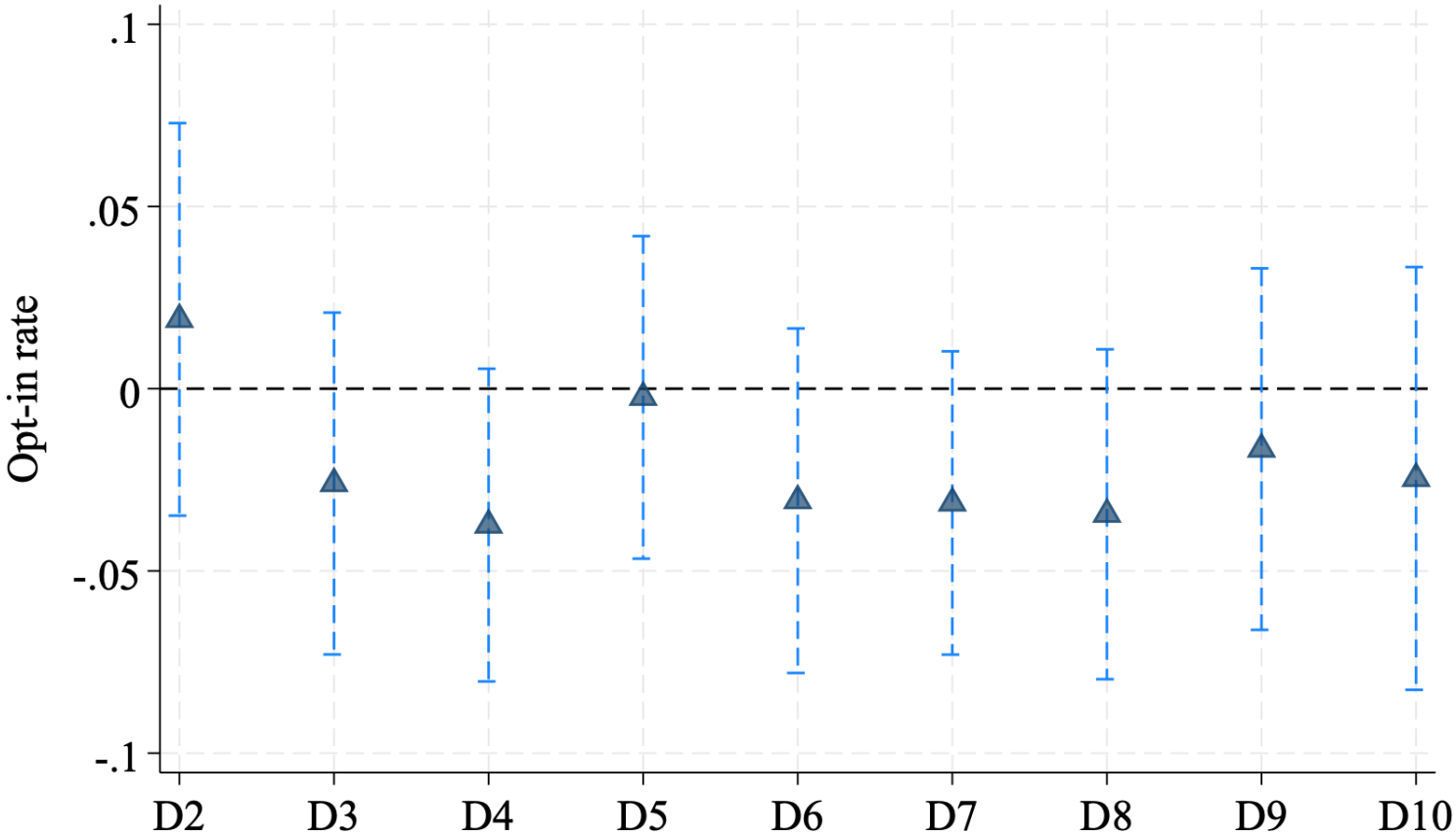
$$y_{isrt} = \alpha_{sr} + \gamma_t + \left(\sum_{k \neq 2012} \beta_k \cdot 1\{t = k\} \cdot T_{sr} \right) + \epsilon_{isrt}$$

Appendix: Opt-Out, Pre and Post



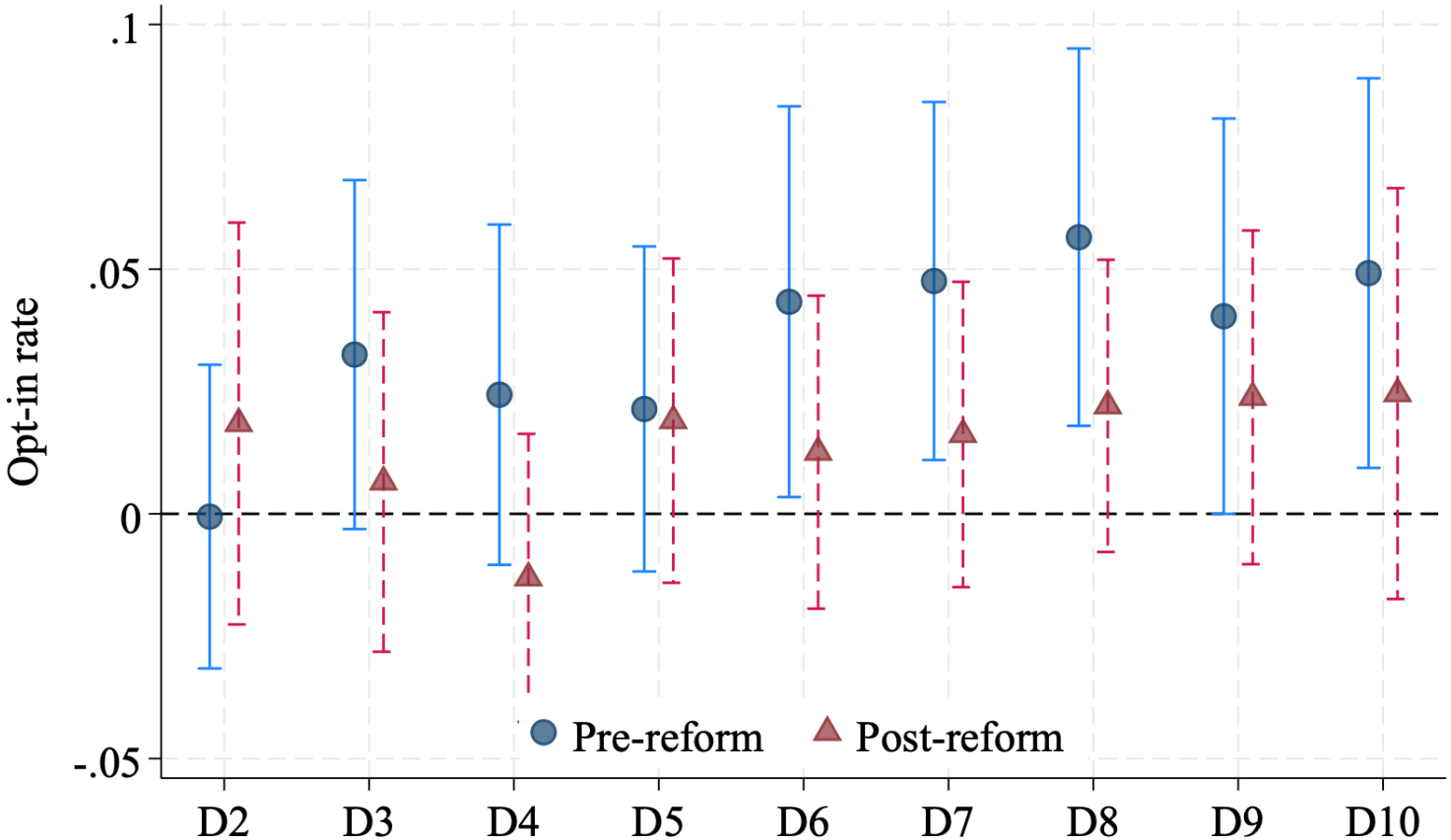
$$y_{isrt} = \alpha_{sr} + \gamma_t + \left(\sum_{k \neq 2012} \beta_k \cdot 1\{t = k\} \cdot T_{sr} \right) + \epsilon_{isrt}$$

Appendix: Opt-In, by Exposure Decile



$$y_{isrt} = \alpha_{sr} + \gamma_t + \left(\sum_{k \neq 2012} \beta_k \cdot 1\{t = k\} \cdot T_{sr} \right) + \epsilon_{isrt}$$

Appendix: Opt-In, Pre and Post



$$y_{isrt} = \alpha_{sr} + \gamma_t + \left(\sum_{k \neq 2012} \beta_k \cdot 1\{t = k\} \cdot T_{sr} \right) + \epsilon_{isrt}$$

Wage Floor Data: By State

- Allows us to define sector-region gap measure, or share of floors below MW:

$$S_{rt} = \frac{1}{N^s_{\cdot}} \sum 1\{W_{srt} < 8.50\}$$

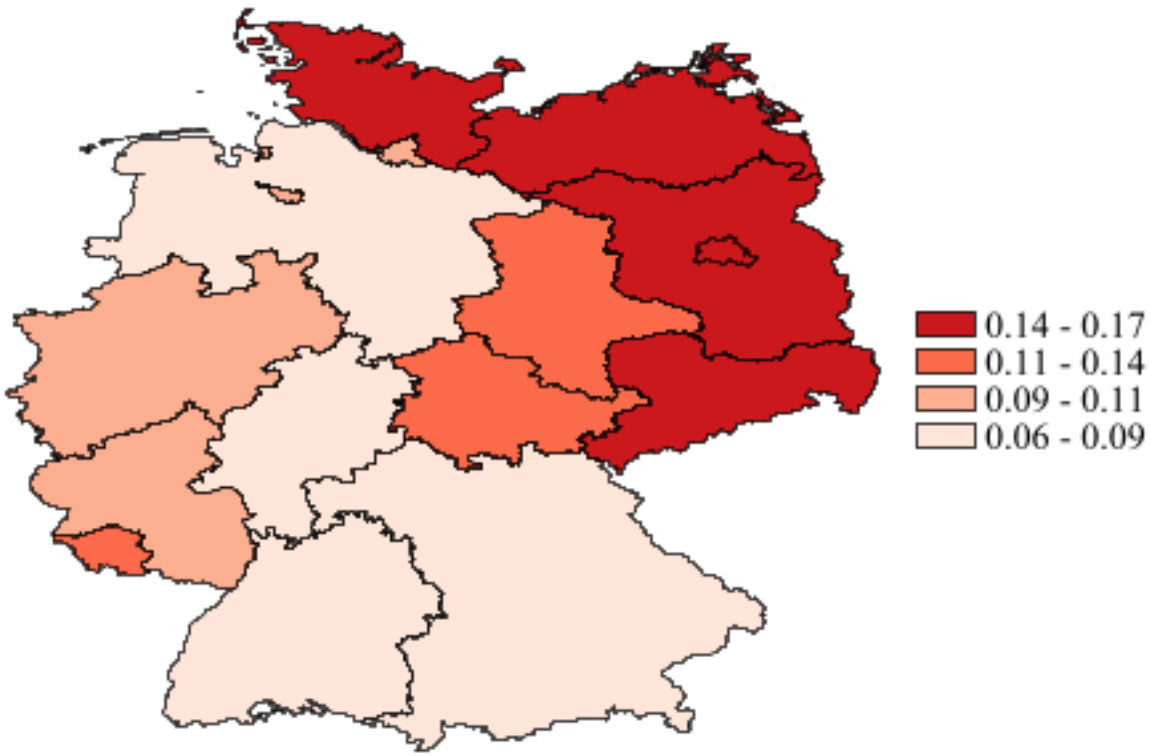


Figure: 2014 Share of Floors Below MW at Federal State Level

Wage Floor Data: By Sector

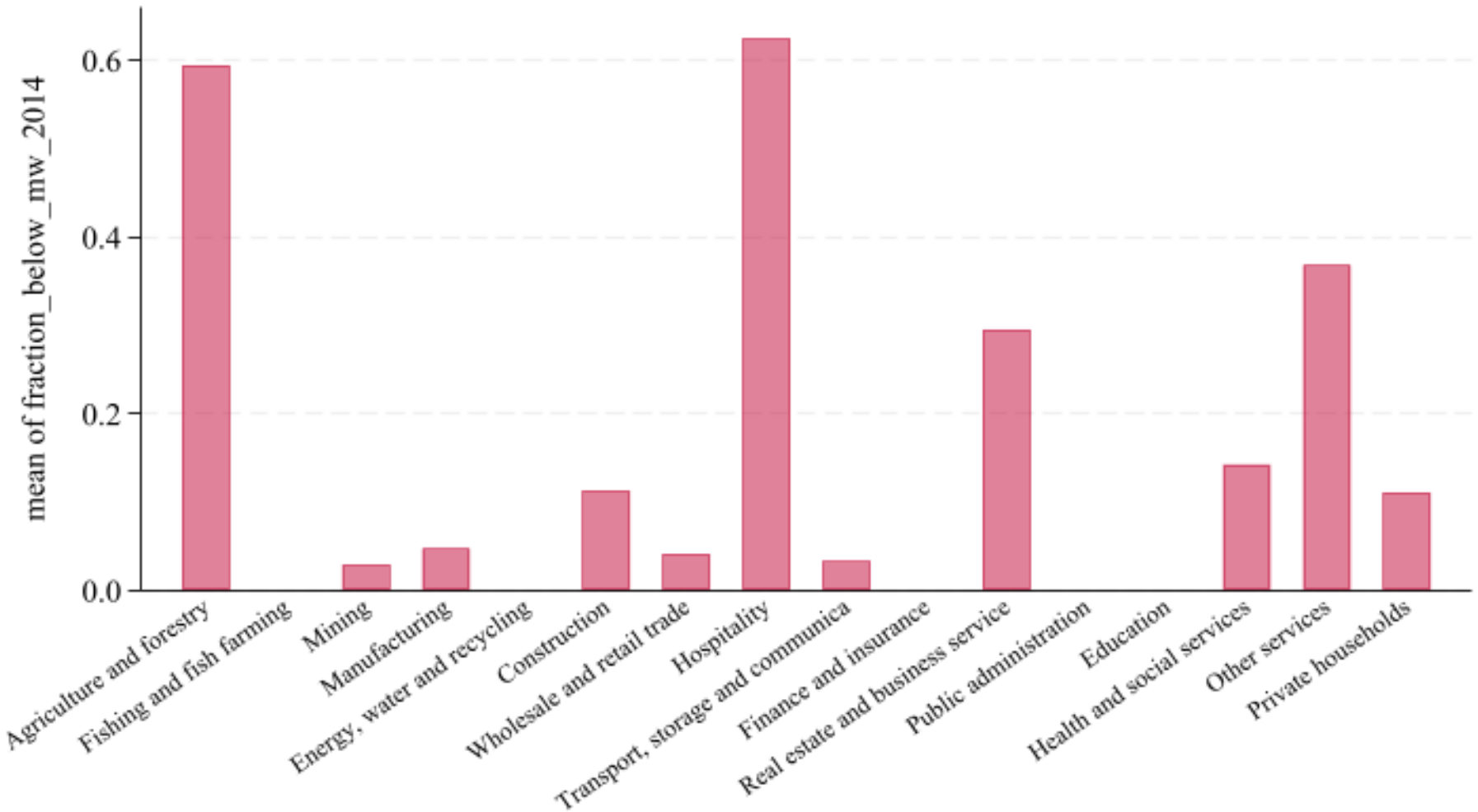


Figure: 2014 Share of Floors Below MW by Sector

Wage Floor Data: Minimum Wage Spikes

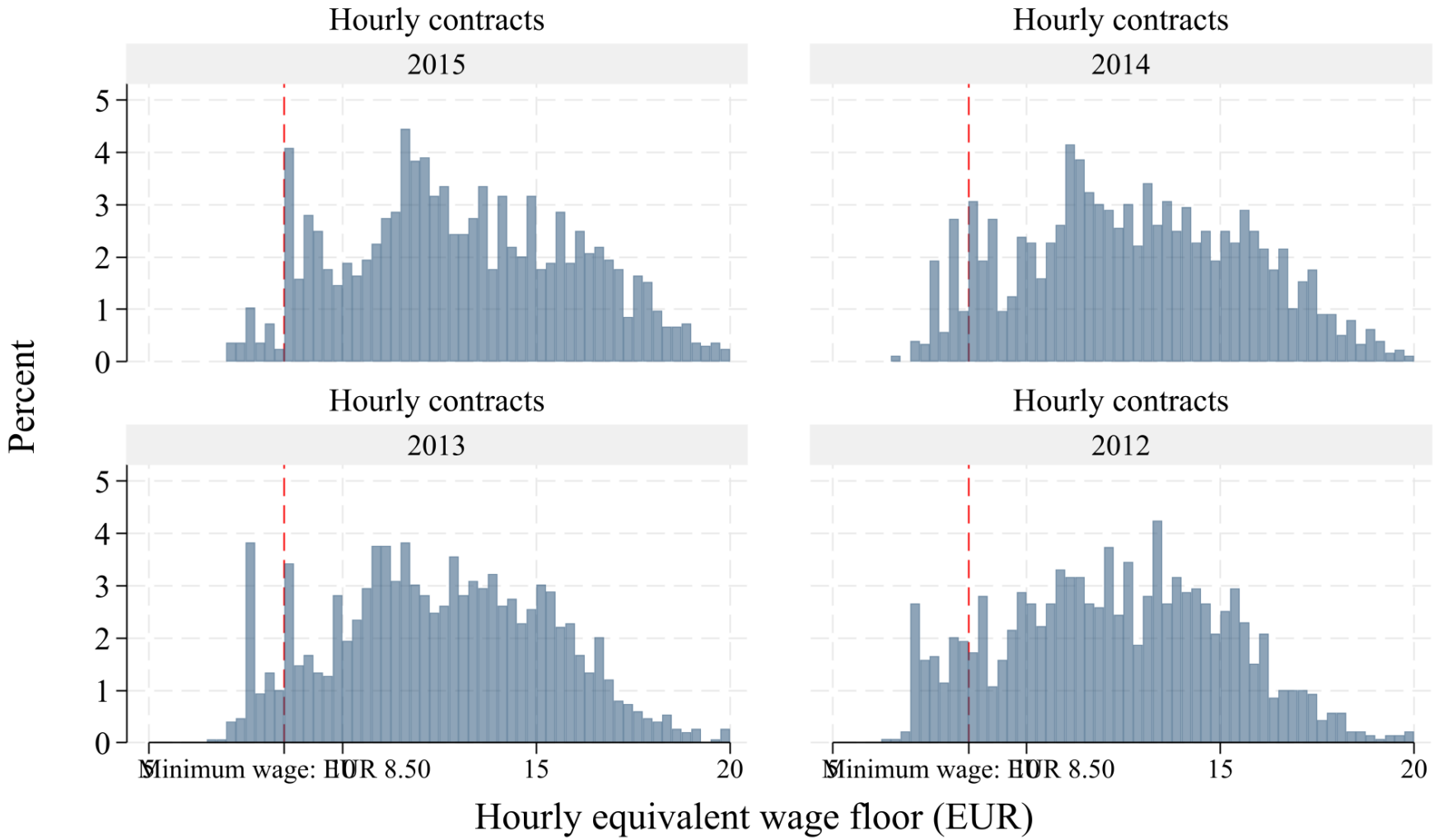


Figure: Wage Floor Histograms

Appendix: Theory: Worker Demand for Coverage

- In keeping with most monopsony models, we take workers' demand for coverage (or higher rewards) as given, and focus on firm supply side
- Alternative view: worker demand for coverage may be higher in more monopsonistic markets, where workers have fewer outside options and are more reliant on coverage to secure higher wages and better amenities
- However, union membership rates in low wage sectors tend to be very low so prospect for effective strikes to maintain coverage in (low-wage) monopsonistic markets is limited
- $\Rightarrow$ focus on firm supply side of coverage decision, and treat worker demand as given

Back to Theory